LVT: Large-Scale Scene Reconstruction via Local View Transformers

Tooba Imtiaz[1,2*], Lucy Chai[1*], Kathryn Heal[1], Xuan Luo[1], Jungyeon Park[1], Jennifer Dy[2], John Flynn[1]

[1] Google [2] Northeastern University

* Denotes Equal Contribution

https://toobaimt.github.io/lvt/

SIGGRAPH ASIA 2025 HONG KONG 香港

Summary

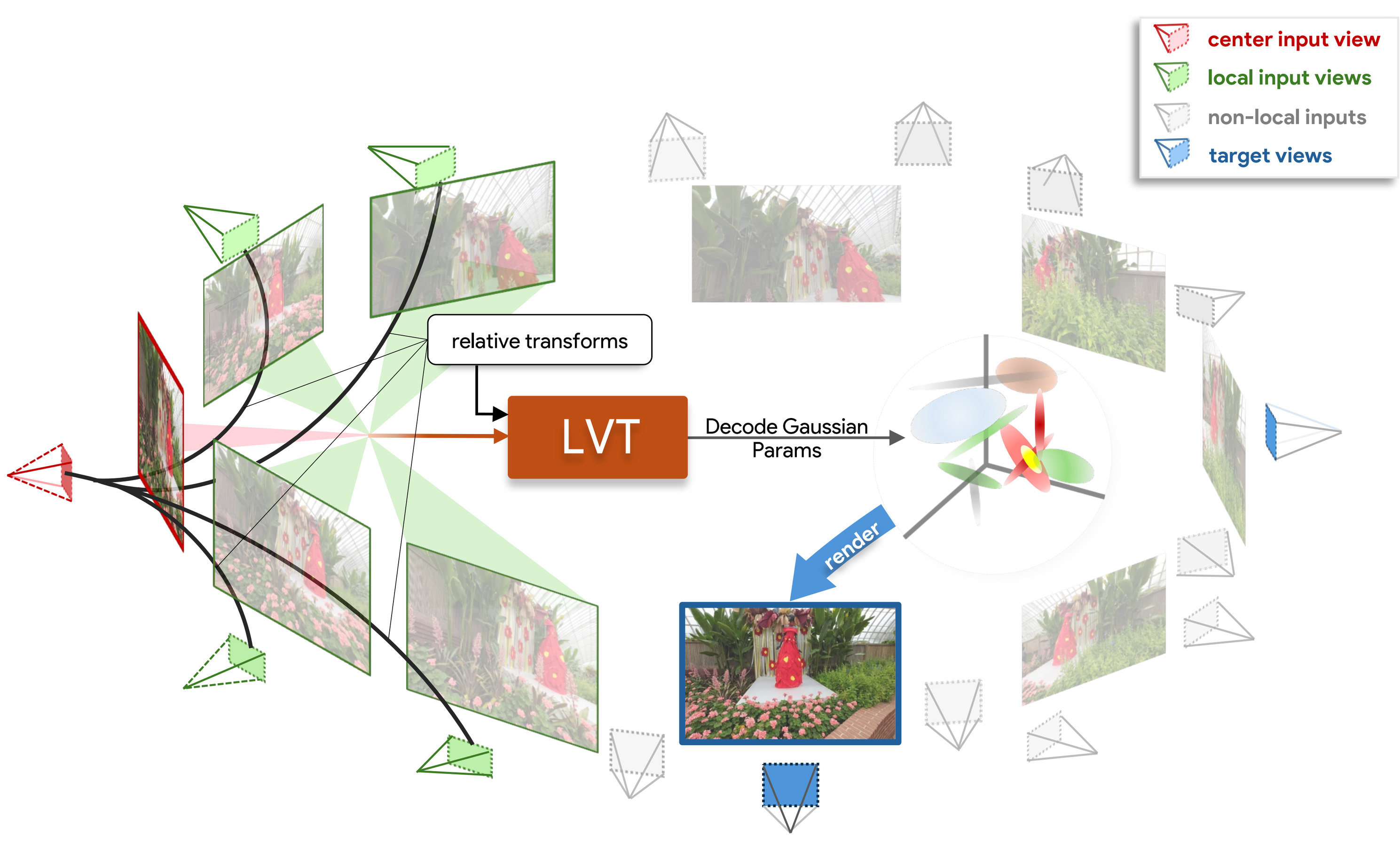
Task: 3D reconstruction of large-scale scenes

- Transformer-based models
  → have quadratic complexity O(N²) with number of input views N
  → unscalable to large, high-resolution scenes
- Per-scene optimization methods (e.g. 3DGS[1])
  → need slow test-time gradient steps

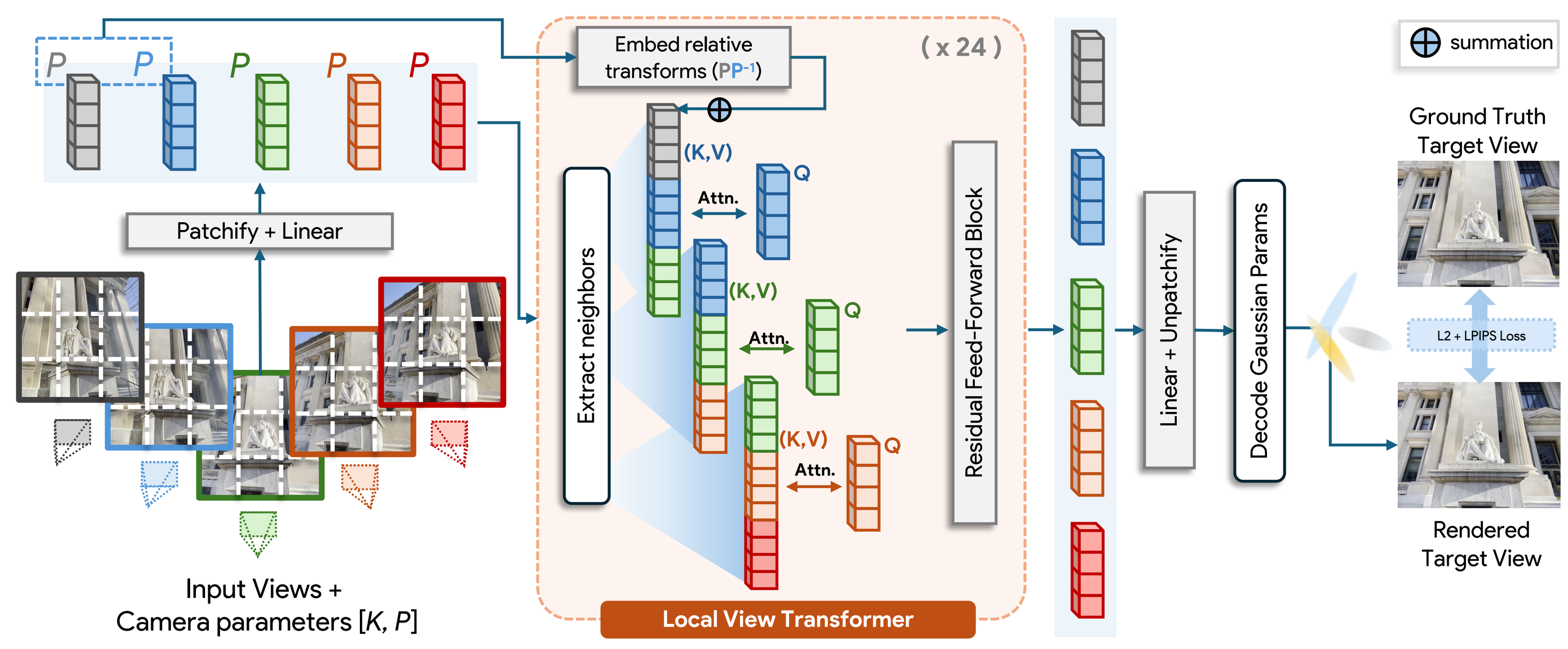
Solution

We propose the Local View Transformer (LVT), an efficient and high-fidelity architecture for large-scale scene reconstruction, which

- Replaces costly global self-attention with a local neighborhood attention
  → scales linearly O(N) with the number of inputs
- Reconstructs arbitrarily large scenes with **state-of-the-art performance in a single feed-forward pass!**



Proposed Method



Local Attention Mechanism

- **Core idea:** Spatially nearby views in an image sequence → most informative for local scene structure
- ~~Full quadratic self-attention~~ → attention between query view and spatial neighbors
- Stacking LVT blocks progressively expands "view receptive field," like CNNs

Transformation Invariant Design

- LVT conditions tokens on relative geometric transformation (pose) between views
- Allows generalizing from short training sequences to arbitrarily long sequences

Scene Representation and View-dependent Opacity

- LVT output is decoded into a 3D Gaussian Splat scene representation
- LVT also introduces view-dependent opacity (using spherical harmonics) to enhance rendering of specular highlights and thin structures

Results

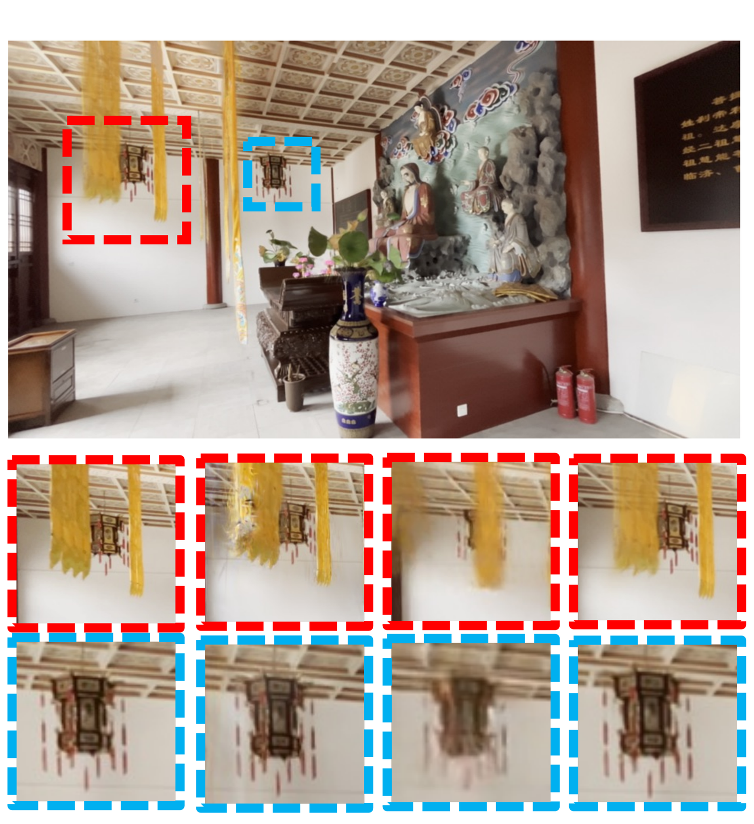
State-of-the-art on DL3DV Benchmark

- LVT outperforms prior SOTA Long-LRM[2] by **+3.54dB PSNR**
- Surpasses per-scene optimized 3DGS[1] by **+2.11dB PSNR**

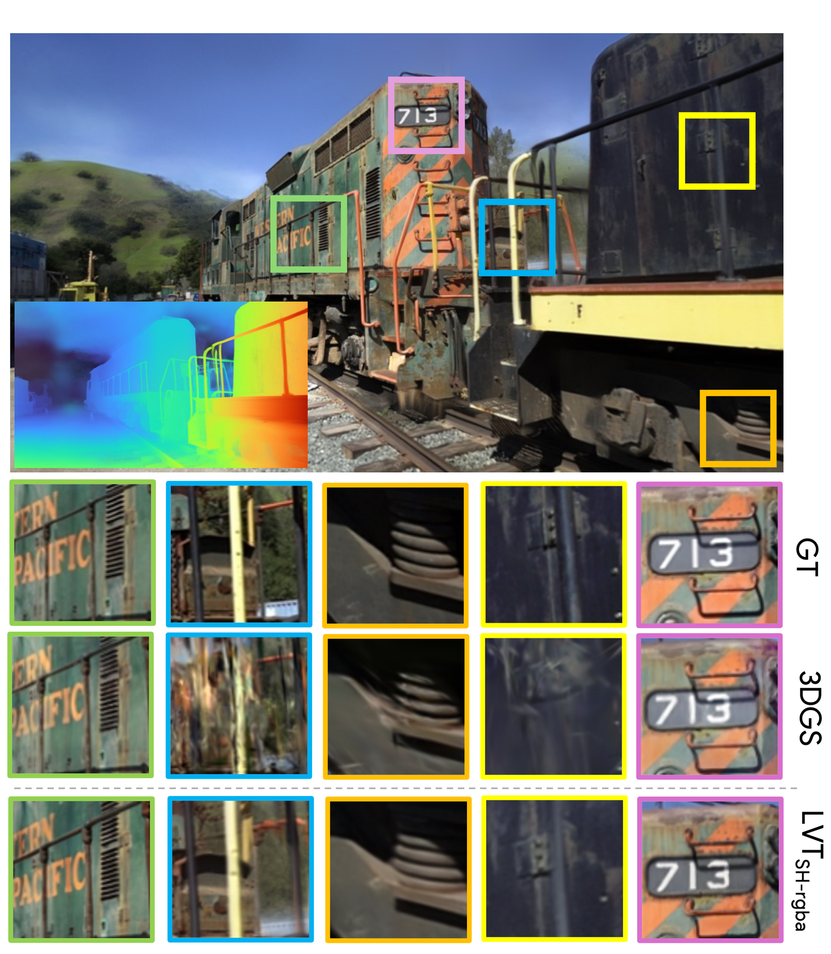
Zero-shot Generalization

- LVT generalizes zero-shot to Tanks&Temples and MipNeRF360
- Outperforms per-scene optimized 3DGS[1] on T&T!

Model	Feed-Forward	DL3DV-140			Tanks&Temples			MipNeRF360		
		PSNR ↑	*SSIM* ↑	*LPIPS* ↓	*PSNR* ↑	*SSIM* ↑	*LPIPS* ↓	*PSNR* ↑	*SSIM* ↑	*LPIPS* ↓
$3DGS_{30k}$	✗	<u>25.53</u>	<u>0.849</u>	<u>0.210</u>	<u>20.45</u>	<u>0.780</u>	<u>0.251</u>	**24.81**	**0.745**	**0.246**
Long-LRM	✓	24.10	0.783	0.254	18.38	0.601	0.363	–	–	–
LVT_{base}	✓	23.65	0.813	0.215	18.33	0.681	0.296	21.51	0.603	0.335
$LVT_{SH-rgba}$	✓	**27.64**	**0.883**	**0.133**	**21.17**	**0.794**	**0.174**	<u>24.53</u>	<u>0.698</u>	<u>0.251</u>



Qualitative Results – DL3DV



Zero-shot evaluation – Tanks&Temples

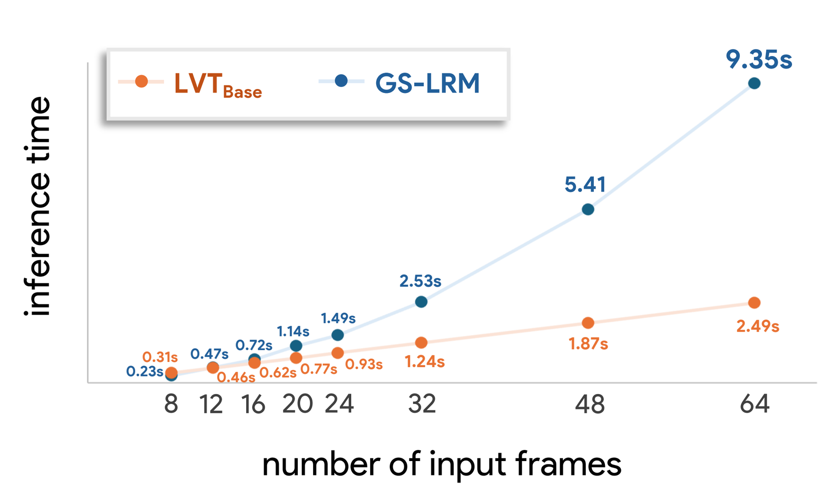
Ablations, Runtime Comparison

Local vs. global attention. When training with 16 views but evaluating with different number of input views, LVT is more robust to varying sequence lengths while also scaling linearly.

Number of Inputs	Complexity	*PSNR* ↑ 12	16	20	32	*SSIM* ↑ 12	16	20	32	*LPIPS* ↓ 12	16	20	32
Global Attention	$O(N^2)$	25.24	25.01	24.39	22.83	0.837	0.830	0.817	0.779	0.133	0.141	0.152	0.182
Local Attention	$O(N)$	**25.32**	**25.12**	**24.69**	**23.31**	**0.846**	**0.842**	**0.834**	**0.807**	**0.130**	**0.135**	**0.141**	**0.167**

Design ablations. Our full model adds mixed-resolution training and regularized spherical harmonics to improve over our initial configuration.

Ablation	Mixed-Res. Train	Spatial Neighbor	Tokenize Cond.	Attention Cond.	SH	σ Reg.	*PSNR* ↑	*SSIM* ↑	*LPIPS* ↓
Initial Config.	✗	✗	Intrinsics	Relative Pose	✗	✗	22.02	0.774	0.196
Mixed-res. Train	✓	✗	Intrinsics	Relative Pose	✗	✗	22.57	0.786	0.186
Add Spatial	✗	✓	Intrinsics	Relative Pose	✗	✗	22.40	0.783	0.190
Conditioning Variants	✗	✗	Plücker	None	✗	✗	20.59	0.696	0.260
	✗	✗	Intrinsics	World Pose	✗	✗	21.44	0.745	0.220
	✗	✗	Intrinsics	Relative Plücker	✗	✗	21.81	0.767	0.202
SH Variants	✗	✗	Intrinsics	Relative Pose	✓	✗	23.89	0.828	0.152
	✗	✗	Intrinsics	Relative Pose	✓	✓	<u>24.69</u>	<u>0.844</u>	<u>0.136</u>
Full Model	✓	✓	Intrinsics	Relative Pose	✓	✓	**27.16**	**0.882**	**0.102**



Runtimes comparison of LVT and global attention. LVT scales linearly, while global attention is quadratic with the number of input views.

References:
[1] "3D Gaussian Splatting for Real-Time Radiance Field Rendering", Kerbl et al. ACM TOG 2023.
[2] "Long-LRM: Long-sequence large reconstruction model for wide-coverage gaussian splats" Ziwen et al. ICCV 2025.
[3] "GS-LRM: Large reconstruction model for 3d gaussian splatting", Zhang et al. ECCV 2024.

Northeastern University Google